

Network Protocols and Service Models

1. Introduction

Network Protocols are a set of **rules and standards** that define how data is transmitted, formatted, and processed across networks.

They allow devices (computers, routers, servers) to **communicate** efficiently and securely.

Service Models define **how data is delivered** between different layers of a network (especially in the OSI and TCP/IP models).

In short:

Protocol = Rules for communication

Service Model = How that communication happens

2. What is a Network Protocol?

A **network protocol** controls how devices exchange data.

It ensures proper communication through defined **syntax**, **semantics**, and **synchronization**.

Example:

When you open a website, your browser uses:

- **HTTP/HTTPS** → to request web pages.
- **TCP/IP** → to transfer data reliably.
- **DNS** → to find the website's IP address.

3. Service Models in Networking

Service models describe **how data is transferred and what guarantees** are provided.

a) Connection-Oriented Service (Reliable)

- Data sent after connection setup.
- Ensures delivery and sequence.
- Example: **TCP** (used in web browsing, file transfer, emails).

b) Connectionless Service (Unreliable)

- Data sent without setup.
- Faster but may lose packets.
- Example: **UDP** (used in live streaming, gaming, VoIP).

4. Network Services

These are functionalities provided by network protocols to users or systems.

Service	Description	Protocols Used
Web Service	Provides website access	HTTP, HTTPS
Email Service	Sending/receiving emails	SMTP, POP3, IMAP
File Sharing	Transfer and download files	FTP, SFTP
Remote Access	Control devices remotely	SSH, Telnet

Name Resolution	Convert names to IPs	DNS
Dynamic IP Assignment	Automatic IP allocation	DHCP

5. Importance of Secure Protocols

- Protects against **data interception** (sniffing).
- Prevents **man-in-the-middle (MITM)** attacks.
- Ensures **data integrity** and **authentication**.
- Builds **user trust** in online communications.

